Lifting a Lipschitz Newton certificate to quantum mechanics

Result, 2026-10-27; written derivation, unrefereed. A classical kick–drift existence certificate for a one-dimensional conservative force with a global Lipschitz constant supplies the model-dependent estimates for a specified noncommutative quantum construction. For every supplied $\hbar > 0$ it gives the Schrödinger representation, an essentially self-adjoint Hamiltonian, a strongly continuous unitary group, and arbitrary-partition convergence of the quantum kick–drift products. On coherent preparations the same classical polygons and quantum products agree at their vertices to $O(\sqrt{\hbar})$, uniformly in the mesh on a fixed time interval. Together with the classical certificate this gives the requested joint observable limit.

This is a **bounded instance**, not a proof of the universal constructive lifting conjecture. The input includes the conservative action and a globally Lipschitz force; the background analytic toolkit includes Nelson's commutator theorem and the graph-domain propagation result cited below. No lower bound on V or derivatives beyond the Lipschitz force are assumed. This real-time theorem does not construct a Euclidean probability measure for a potential unbounded below.

The loss of the earlier $O(\hbar)$ rate is necessary at this level of regularity: $V(q) = q|q|/2$ has globally Lipschitz force, but a centered coherent packet has a momentum error of exact order $\sqrt{\hbar}$ at small fixed positive times. No claim of literature priority is made.

1. The input certificate and quantum target

Let $m > 0$, $V \in C^1(\mathbb{R};\mathbb{R})$, and suppose

$$|V'(x) - V'(y)| \le L|x - y|, \qquad L \ge 0. \tag{1}$$

Thus $V \in C^{1,1}$ and V'' exists almost everywhere with $|V''| \le L$. The input is the mechanical action with kinetic term $p^2/(2m)$ and potential V, the initial point $z_0 = (q_0, p_0)$, and the classical kick–drift polygons

$$p^{\mathrm{cl}}_{j+1} = p^{\mathrm{cl}}_j - s_j V'(q^{\mathrm{cl}}_j), \qquad q^{\mathrm{cl}}_{j+1} = q^{\mathrm{cl}}_j + s_j p^{\mathrm{cl}}_{j+1}/m. \tag{2}$$

Take their linear interpolation for a partition π of $[0, T]$. The usual Lipschitz certificate supplies a unique limit $z(t)$ and

$$\sup_{t \le T} \|z^{\mathrm{cl}}_\pi(t) - z(t)\|_\omega \le R^{\mathrm{cl}}_T |\pi|, \qquad \|(q,p)\|_\omega = |q| + \frac{|p|}{m\omega}. \tag{3}$$

For example, the proof in the ODE lesson uses a common bounded region, force-balance residual and Gronwall stability. Here $\omega > 0$ is a fixed auxiliary unit; R^{cl}_T depends only on the classical data and T. No quantum convergence is included in the certificate.

For each $\hbar > 0$ set, on $L^2(\mathbb{R})$,

$$\begin{gathered} Qf(q) = qf(q), \quad P = -i\hbar\partial_q, \quad H_\hbar = P^2/(2m) + V(Q), \\ K_{\hbar,s} = e^{-isV(Q)/\hbar}, \quad D_{\hbar,s} = e^{-isP^2/(2m\hbar)}. \end{gathered} \tag{4}$$

The self-adjoint Q, P obey the Weyl relations and $[Q, P] = i\hbar I$ on Schwartz space. Unlike the Koopman construction, these observables are noncommutative.

2. The same force bound controls the operator domains

Fix $\hbar > 0$ and calibrated units and put $N = 1 + Q^2 + P^2$. The input (1) implies

$$|V'(q)| \le |V'(0)| + L|q|, \qquad |V(q)| \le |V(0)| + |V'(0)||q| + \tfrac{L}{2}q^2. \tag{5}$$

Using the harmonic-oscillator graph bounds for Q^2, P^2, QP, PQ, these estimates and the almost-everywhere bound on V'' imply on $\mathcal{S}(\mathbb{R})$

$$\|H_\hbar f\| \le C_\hbar \|Nf\|, \qquad \|[H_\hbar, N]f\| \le C_\hbar \|Nf\|. \tag{6}$$

The commutator is a weak-derivative identity:

$$[H_\hbar, N] = -\frac{i\hbar}{m}(PQ + QP) + i\hbar(V'(Q)P + PV'(Q)). \tag{7}$$

Products with V' have the weak derivative V''; no third derivative is used. For $f \in \mathcal{S}$, $H_\hbar f \in D(N)$, so (6) gives

$$|\langle H_\hbar f, N^2 f\rangle - \langle N^2 f, H_\hbar f\rangle| \le 2C_\hbar \langle f, N^2 f\rangle. \tag{8}$$

Schwartz space is a core for N^2. Nelson's commutator theorem therefore makes $H_\hbar$ essentially self-adjoint on $\mathcal{S}$. With $G = N^2 - \inf \sigma(N^2)$, the same estimate and Energy-Limited Quantum Dynamics, Theorems 3.7–3.8 (statements and hypotheses read) show that $U_\hbar(t) = e^{-itH_\hbar/\hbar}$ preserves $D(N)$ and is strongly continuous in its N-graph norm. The constants in (6)–(8) follow from $m, \hbar, L, V(0), V'(0)$; they are not additional assumptions on the mechanical model.

3. A fixed-action quantum product estimate

The force certificate also controls the commutator of the two factors. Write $T_\hbar = P^2/(2m)$ and, initially on Schwartz space,

$$[T_\hbar, V(Q)] = -\frac{i\hbar}{2m}(PV'(Q) + V'(Q)P). \tag{9}$$

It extends as a bounded map $D(N) \to L^2$. To justify the product argument without assuming higher derivatives, mollify V with an even compactly supported unit-mass mollifier. The smooth V_n satisfy $\|V_n''\|_\infty \le L$ and have uniformly linearly growing first derivatives. Moreover, Taylor's inequality gives

$$\|V_n - V\|_\infty \le \tfrac{1}{2} L\, \varepsilon_n^2 \int y^2 \rho(y)\, dy \longrightarrow 0. \tag{10}$$

This uniform bound holds even when V is unbounded below.

For these V_n, define the finite constants

$$\eta_T = \hbar^{-1}\|[T_\hbar, N]N^{-1}\|, \quad \eta_V = \sup_n \hbar^{-1}\|[V_n(Q), N]N^{-1}\|, \quad B = \sup_n \|[T_\hbar, V_n(Q)]N^{-1}\|. \tag{11}$$

They are finite by (5) and the graph bounds above; their values can be bounded from the classical data $m, L, V'(0)$ and the chosen $\hbar$ and units. The N-graph norms of the kinetic, potential and full groups are bounded by $e^{\eta_T|t|}$, $e^{\eta_V|t|}$, and $e^{(\eta_T+\eta_V)|t|}$, respectively.

For smooth V_n, differentiation of $U_n(s-r)D_r K_{n,r}$ and a second Duhamel integration yield, on Schwartz vectors,

$$\|(D_s K_{n,s} - U_n(s))f\| \le \frac{Bs^2}{2\hbar^2} e^{(\eta_T+\eta_V)s}\|Nf\|. \tag{12}$$

The left factors in the integral are unitary; the commutator (9) acts on $D_u K_{n,r} f$. Density extends (12) to $D(N)$. Telescoping along the exact U_n orbit and using the graph bound gives

$$\|U_{n,\pi}(T)f - U_n(T)f\| \le \frac{BT\, e^{(\eta_T+\eta_V)T}}{2\hbar^2}|\pi|\, \|Nf\|, \qquad f \in D(N). \tag{13}$$

Equation (10) makes $U_n(t) \to U_\hbar(t)$ and each finite product $U_{n,\pi}(T) \to U_{\hbar,\pi}(T)$ in operator norm for fixed $\hbar, T, \pi$, by the bounded-perturbation Duhamel estimate. The constants in (13) are uniform in n, so (13) passes to V. It proves arbitrary-partition strong convergence for $f \in D(N)$; density and unitarity extend convergence to all L^2 vectors. In particular, the conjecture note's fixed-$\hbar$ Cauchy criterion holds, with the explicit graph-domain rate in (13). The factor $\hbar^{-2}B$ need not remain bounded as $\hbar \downarrow 0$.

4. The classical and quantum polygons

Use the coherent packet of the quantum-blocks note, equation (16), and let (q_j, p_j) be the means after the first j quantum cells. The exact identities

$$K_s^* P K_s = P - sV'(Q), \qquad D_s^* Q D_s = Q + sP/m \tag{14}$$

hold on the propagated states; the $C^{1,1}$ potential preserves the needed graph domain. Define

$$W_j = \operatorname{sd} Q_j + \frac{\operatorname{sd} P_j}{m\omega}, \quad C = \sqrt{\frac{2}{m\omega}}, \quad \beta = \omega + \frac{L}{m\omega}. \tag{15}$$

As in the earlier note, (1) and (14) give

$$W_j \le C\sqrt{\hbar}\, e^{\beta t_j}. \tag{16}$$

No higher potential derivative or assumption of Gaussian shape after the first step is used.

Put $e_j = \|(q_j, p_j) - (q_j^{\mathrm{cl}}, p_j^{\mathrm{cl}})\|_\omega$. For each kick, $|\langle V'(Q)\rangle - V'(q_j^{\mathrm{cl}})| \le L(\operatorname{sd} Q_j + |q_j - q_j^{\mathrm{cl}}|)$. The subsequent drift yields

$$e_{j+1} \le (1 + s_j\omega)\left(1 + \frac{s_j L}{m\omega}\right) e_j + \frac{s_j L}{m\omega}(1 + s_j\omega) W_j. \tag{17}$$

Since $e_0 = 0$, (16) and $1 + x \le e^x$ imply the **mesh-uniform** bound

$$e_j \le \frac{LC}{m\omega}\, t_j e^{\beta t_j}\sqrt{\hbar}. \tag{18}$$

Combining this with the classical certificate (3) gives

$$\|(q_j, p_j) - z(t_j)\|_\omega \le \frac{LC}{m\omega} T e^{\beta T}\sqrt{\hbar} + R_T^{\mathrm{cl}}|\pi|. \tag{19}$$

For $b \in C_c^\infty(\mathbb{R}^2)$, the Fourier/Weyl argument in the groupoid note, before its second-order refinement, bounds the expectation error by a symbol-dependent constant times the sum of W_j and (19). Therefore the observable limit required by the conjecture holds at all partition vertices and along every joint schedule $\hbar, |\pi| \to 0$.

The proof uses the classical **input bound** (1) for both the graph and packet estimates, while the certificate's polygon error (3) controls recovery of the classical flow. The graph estimate (13) is a separate fixed-$\hbar$ quantum estimate derived using the declared operator toolkit; it is not assumed in (3).

5. Why the square-root rate cannot generally be improved

Set $m = 1$ and

$$V(q) = \tfrac{1}{2} q|q|, \qquad V'(q) = |q|. \tag{20}$$

The force is globally Lipschitz, and the classical solution from $(0, 0)$ is uniquely stationary. Define the unitary dilation $(D_\hbar f)(q) = \hbar^{-1/4} f(q/\sqrt{\hbar})$. Homogeneity gives

$$D_\hbar^{-1} H_\hbar D_\hbar = \hbar H_1, \qquad D_\hbar^{-1} P_\hbar D_\hbar = \sqrt{\hbar}\, P_1. \tag{21}$$

For the centered Gaussian $D_\hbar g_1$, the exact quantum momentum is therefore $\langle P_\hbar\rangle_t = \sqrt{\hbar}\, f(t)$, where $f(t) = \langle P_1\rangle_t$. Ehrenfest's identity gives

$$f'(0) = -\langle |Q|\rangle_{g_1} = -1/\sqrt{\pi}. \tag{22}$$

Thus $f(t) \neq 0$ for every sufficiently small fixed $t > 0$. The mean error is exactly a nonzero multiple of $\sqrt{\hbar}$ there. This counterexample does not contradict the $O(\hbar)$ theorem for bounded V''': V' here has a corner at the origin.

6. Consequence for STATE

The bounded certificate-to-quantum test is affirmative for the specified one-dimensional conservative $C^{1,1}$ action class. It constructs the noncommutative target and a quantum product estimate from the classical force data and the named analytic toolkit, and uses the classical polygon certificate in the observable limit. The universal conjecture, singular classical actions, quantum completion uniqueness and independent positive-action necessity remain open.